
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ
IN THE NAME OF ALLAH

FINAL YEAR PROJECT | CDRB ADJUDICATION

SDR-Based Doppler Radar

using SDR for Velocity Measurement

GROUP MEMBERS:

S/LT BILAL PN

S/LT SANAULLAH PN

S/LT FARHAN PN

S/LT HUZAIFA PN

Advisor:

ASSISTANT PROFESSOR MOHAMMAD MUSTAQIM

Introduction

- Radar systems are fundamental tools for detecting object motion and measuring velocity.
- Traditional hardware radars are expensive, fixed-frequency, and difficult to reconfigure.
- Software Defined Radio (SDR) replaces dedicated hardware with flexible, software-controlled signal processing.
- The Analog Devices ADALM-PLUTO (PlutoSDR) is a low-cost, portable SDR platform operating from 325 MHz – 3.8 GHz.
- This project implements a complete Doppler radar system on SDR using GNU Radio for real-time velocity estimation.

Problem Statement

Conventional radar systems for velocity measurement are costly and hardware-dependent; a flexible, low-cost SDR-based Doppler radar is needed for accurate real-time velocity measurement without dedicated RF hardware.

Cost

Dedicated radar ICs cost thousands of dollars

Inflexibility

Fixed-frequency hardware cannot be reprogrammed

Complexity

Separate TX/RX chains increase system complexity

Proposed Solution

Use SDR as a low-cost, software-reconfigurable platform to transmit a CW signal and capture the Doppler-shifted echo for real-time velocity computation via GNU Radio.

Low-Cost Hardware

- SDR
- No custom RF PCB needed
- USB-powered portable device

Software-Defined Processing

- GNU Radio
- Real-time signal processing
- Reconfigurable frequency & gain

Doppler Velocity Engine

- CW signal @ 2.482 GHz
- FFT-based frequency shift detection
- Velocity computed from fd = 2v·fc/c

Doppler Effect – Principle

$$fd = 2 \cdot v \cdot fc / c$$

fd → Doppler frequency shift (Hz)

v → Target radial velocity (m/s)

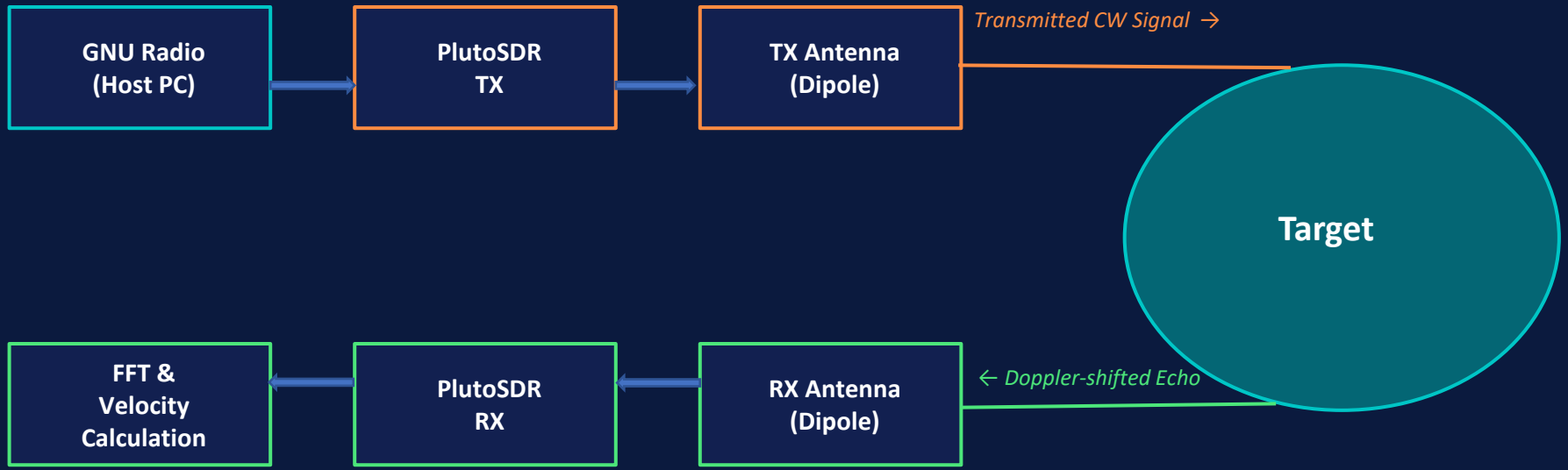
fc → Carrier frequency (Hz)

c → Speed of light = 3×10^8 m/s

How It Works

- Radar transmits a continuous wave (CW) at fc = 2.482 GHz
- Moving target reflects the signal — reflected wave has a shifted frequency
- Frequency difference fd is proportional to velocity v
- Approaching targets → positive fd | Receding targets → negative fd

System Block Diagram



Signal Processing Flow

01

CW Generation

GNU Radio generates a 2.482 GHz CW tone via the PlutoSDR TX

02

Signal Transmission

Antenna radiates the signal; wave propagates to moving target

03

Echo Reception

SDR RX captures reflected signal

04

Mixing / Demodulation

Received signal is mixed to 0 Hz and decimated to 8 kS/s; Doppler shift appears as a low-frequency tone

05

FFT Analysis

2048-point FFT, clutter map and peak detection

06

Velocity Computation

$v = fd \times c / (2 \times fc)$ → shown live in GNU Radio and on the LED display

SDR Platform – PLUTO (PlutoSDR)

Parameter	Specification
RF Frequency Range	325 MHz – 3.8 GHz
Bandwidth	Up to 20 MHz
ADC / DAC Resolution	12-bit
Interface	USB 2.0 (High-Speed)
Transceiver IC	Analog Devices AD9363
Carrier Freq (Used)	2.482 GHz (ISM band)
TX Power	Up to +7 dBm
GNU Radio Support	gr-iio PlutoSDR blocks

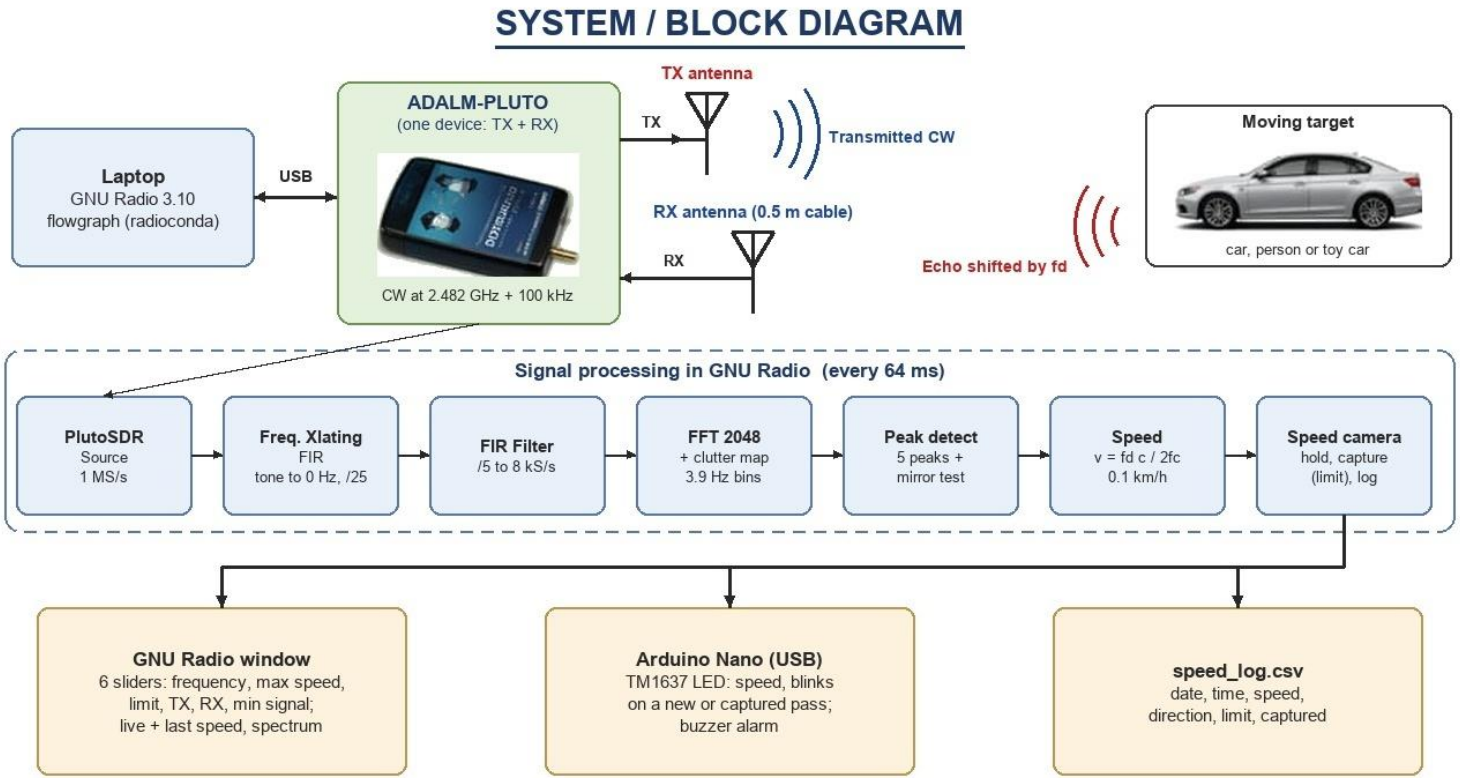
Why PlutoSDR?

- Full-duplex transceiver — simultaneous TX and RX
- Wide tuning range — suitable for multiple radar bands
- Native GNU Radio support (gr-iio)
- Low cost — ideal for academic prototyping
- Compact form factor — portable deployment

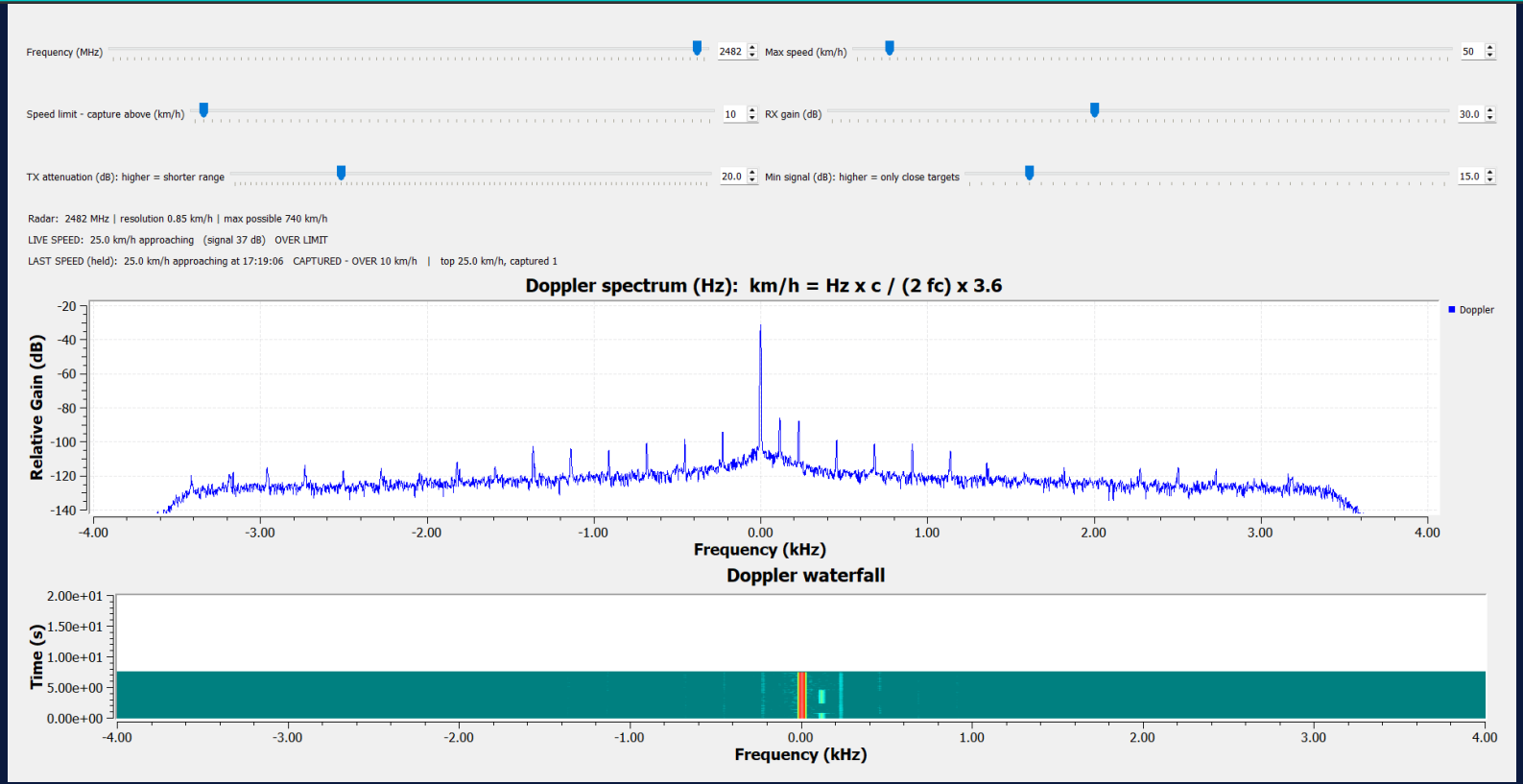
System Specifications

Parameter	Value	Note
Carrier Frequency	2.482 GHz	ISM band; adjustable 2400–2483 MHz
Waveform Type	CW (Continuous Wave)	Simplest Doppler radar waveform
Sampling Rate	1 MSPS	Decimated to 8 kS/s for Doppler
FFT Size	2048 points	0.26 s window, 3.9 Hz bins
Doppler Resolution	0.85 km/h per bin	0.1 km/h readout (peak interpolation)
Max Measurable Velocity	50 km/h (default)	Adjustable up to 740 km/h (slider)
Antenna Type	Dipole (2.4 GHz)	~2 dBi gain, omnidirectional
TX-RX Separation	≥ 0.5 m	Reduce self-interference
Processing Platform	GNU Radio 3.10	gr-iio PlutoSDR blocks + Python

Hardware Setup & System Diagram



GNU Radio Implementation



Test Results

Test	Result	Details
Speed accuracy	Error below 0.05 km/h	Test targets 5–700 km/h injected through the Pluto
Direction	Always correct	Approaching (+) and receding (−)
Response time	0.15 s live, 0.75 s held	Held speed on the LED after the target leaves
Empty room	No false detections	90 s with nobody moving
Walking person	Detected, 3–13 km/h	Signal 24–47 dB above the noise
Toy car	Detected at close range	3–17 km/h, driven straight at the antennas
Speed camera	Captured above the limit	Logged to CSV; LED keeps blinking

TENTATIVE BUDGET

COMPONENTS	COST
Pluto SDR Device	85k-90k pkr
Tx/Rx Antennas (2 units)	5k-8k pkr
RF Components (cables, connectors)	2k-4k pkr
Processing system (laptop)	00 pkr (existing)
Software tools (GNU Radio)	00 pkr (free for student)
Miscellaneous	4k-5k pkr (Arduino Nano, LED display, buzzer)
Total	96k-107k pkr

GANTT CHART - PROJECT TIMELINE



CONCLUSION

A fully functional SDR-based Doppler radar — built and validated.

- ✅ Successfully implemented Doppler radar using PlutoSDR — no custom RF hardware required
- ✅ The system measures velocity from the Doppler shift in real time in GNU Radio and works as a speed camera with an LED display and adjustable frequency and speed limit
- ✅ The project demonstrate practical use of software defined radio in modern radar application with scope for future improvements such as range detection and higher accuracy


THANK
YOU